Notes on Axionic wormholes

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1 Action principle and equations of motion

Our starting point is the following action

$$\mathcal{S} = \int d^D x \sqrt{g} \left[\frac{1}{2\kappa} \left(-\mathcal{R} + 2\Lambda \right) + \frac{1}{2(D-1)!} H_{\mu\ldots\nu} H^{\mu\ldots\nu} \right] . \tag{1.1}$$

H is a $(D-1)$-form field strength satisfying the Bianchi identity $dH = 0$. Thus, one can always write locally $H = dB$ where B is a $(D-2)$-form field. The equations of motion are simply $d\star H = 0$. We can see classically that the dynamics of H is locally governed by a pseudo-scalar field χ defined as $\star H = i\, d\chi$.

That holds at the quantum level. To see this we introduce χ as a Lagrangian multiplier in the following 1st class formalism action

$$\begin{aligned} \mathcal{S}_{1st} &= \int d^D x \frac{\sqrt{g}}{2\kappa} \left(-\mathcal{R} + 2\Lambda \right) + \frac{1}{2} \int H \wedge \star H + \int \chi \wedge dH \\ &= \int d^D x \frac{\sqrt{g}}{2\kappa} \left(-\mathcal{R} + 2\Lambda \right) + \frac{1}{2} \int H \wedge \star H - \int d\chi \wedge H + \mathcal{S}_{\partial M} . \end{aligned} \tag{1.2}$$

where

$$\mathcal{S}_{\partial M} = \int_{\partial M} \chi \wedge H . \tag{1.3}$$

The variational principle is well defined if we impose Dirichlet boundary conditions for χ and for H. The path-integral whose action is defined by (1.1) can be recovered by integrating out χ imposing Bianchi's identity. In the meantime, the action is quadratic in H so we can integrate out H as well by solving its equation of motion

$$H = \star\, d\chi \Rightarrow \star H = (-1)^{D-1} d\chi . \tag{1.4}$$

Indeed, inserting back into the action we get

$$\mathcal{S}_\chi = \int d^D x \frac{\sqrt{g}}{2\kappa} \left(-\mathcal{R} + 2\Lambda \right) - \frac{1}{2} \int d\chi \wedge \star d\chi - \int_{\partial M} \chi \wedge \star d\chi . \tag{1.5}$$

Instanton configurations have the effect of modifying the action by mean of a non-perturbative potential $V_{\rm np}(\chi)$ so that the total action in the pseudo-scalar variable reads

$$\mathcal{S} = \int d^D x \sqrt{g} \left[\frac{1}{2\kappa} \left(-\mathcal{R} + 2\Lambda \right) - \frac{1}{2} \partial_\mu \chi \partial^\mu \chi + V_{\rm np}(\chi) \right] + \mathcal{S}_{GHY} + S_{\partial M} . \tag{1.6}$$

We look for homogeneous and isotropic solutions of the form

$$ds^2 = N^2(\tau) d\tau^2 + a^2(\tau) d\Sigma_d^2 \quad , \quad \chi = \chi(\tau) . \tag{1.7}$$

We obtain the following expressions for the determinant of the metric and the Ricci scalar

$$\sqrt{g} = N a^d \quad , \quad \mathcal{R} = -\frac{2d}{N} \frac{d}{d\tau} \left(\frac{a'}{aN} \right) + d(d-1) \frac{k}{a^2} - d(d+1) \left(\frac{a'}{aN} \right)^2 . \tag{1.8}$$

Finally, let's also mention the form of the Gibbons-Hawking term

$$\mathcal{S}_{GHY} = -\frac{1}{\kappa}\int_{\partial M} d^d x\sqrt{h}K = -\frac{1}{\kappa}\frac{\partial_\tau}{N}V_d(a^d) = -\frac{dV_d a' a^{d-1}}{\kappa N}\,. \tag{1.9}$$

Hence the Lagrangian density of the Einstein-Hilbert part of the action reads

$$\begin{aligned}\mathcal{L}_{EH} &= -\frac{Na^d}{2\kappa}\left[-\frac{2d}{N}\frac{d}{d\tau}\left(\frac{a'}{aN}\right) + d(d-1)\frac{k}{a^2} - d(d+1)\left(\frac{a'}{aN}\right)^2 + 2\Lambda\right] \\ &= -\frac{1}{2\kappa}\left[\frac{d(d-1)(a')^2 a^{d-2}}{N} + d(d-1)Nka^{d-2} - 2N\Lambda a^d\right] + \frac{d}{d\tau}\left[\frac{da^{d-1}a'}{\kappa N}\right]\,.\end{aligned} \tag{1.10}$$

The boundary term coming from this total derivative will get exactly canceled by the Gibbons-Hawking term (see (1.9)). We also have

$$\mathcal{L}_\chi = -\frac{a^d}{2N}(\partial_\tau\chi)^2 + Na^d V_{\rm np}(\chi)\,. \tag{1.11}$$

Also, note that taking the trace of Einstein equation allows one to evaluate the on-shell action as a function of the on-shell solution χ. Indeed,

$$\begin{aligned} G = \kappa T^\chi - \Lambda D \Rightarrow \mathcal{R}\left(1-\frac{D}{2}\right) &= \kappa T^\chi - \Lambda D\,, \\ T^\chi_{\mu\nu} = -\partial_\mu\chi\partial_\nu\chi - g_{\mu\nu}\left(-\frac{1}{2}(\partial_\lambda\chi)^2 + V_{\rm np}(\chi)\right) &\Rightarrow T^\chi = \left(\frac{D}{2}-1\right)(\partial_\lambda\chi)^2 - DV_{\rm np}(\chi)\,.\end{aligned} \tag{1.12}$$

After some manimpulations one obtains

$$\frac{-\mathcal{R}+2\Lambda}{2\kappa} = \frac{1}{2}(\partial_\lambda\chi)^2 - \frac{DV_{\rm np}}{D-2} - \frac{2\Lambda}{\kappa(D-2)}\,. \tag{1.13}$$

We also have the additional boundary terms

$$\mathcal{S}_{GHY} = -\frac{dV_d a' a^{d-1}}{\kappa N}\bigg|_{\partial M}\,, \quad \mathcal{S}_{\partial M} = -\frac{V_d}{N}\,a^d\chi\partial_\tau\chi\bigg|_{\partial M}\,. \tag{1.14}$$

The on-shell action thus takes the simple expression

$$S^{\rm on-shell} = V_d\int d\tau N a^d\left(-\frac{2V_{\rm np}}{D-2} - \frac{2\Lambda}{\kappa(D-2)}\right) + \mathcal{S}_{GHY} + \mathcal{S}_{\partial M}\,. \tag{1.15}$$

This on-shell action is an important quantity since the final question one should answer is how it evolves as a function of the number of bubbles. The flow paramater is $u = N\tau$. Remark that the on-shell action does not depend on the kinitic term and can be easily deduced from this formula once χ and the scale factor is known.

Let's now derive the equation of motions. The simplest one is obtain by varying wrt to χ yielding

$$\partial_\tau\left(\frac{a^d}{N}\partial_\tau\chi\right) + Na^d\partial_\chi V_{np}(\chi) = 0\,. \tag{1.16}$$

Varying with respect to a in $\mathcal{L}_{EH} + \mathcal{L}_\chi$ yields

$$\begin{aligned}&\Bigg[-\frac{da^{d-1}}{2N}(\partial_\tau\chi)^2 + dNa^{d-1}V_{\rm np}(\chi) - \frac{1}{2\kappa}\Big(\frac{d(d-1)(d-2)}{N}a'^2a^{d-3} \\ &+ d(d-1)(d-2)kNa^{d-3} - 2N\Lambda da^{d-1}\Big)\Bigg]\delta a - \frac{1}{2\kappa}\frac{d(d-1)2a'a^{d-2}}{N}\delta(a')\,.\end{aligned} \tag{1.17}$$

Thus, integrating by parts the second term

$$\begin{aligned}&\Bigg[-\frac{da^{d-1}}{2N}(\partial_\tau\chi)^2+dNa^{d-1}V_{\rm np}(\chi)-\frac{1}{2\kappa}\Bigg(\frac{d(d-1)(d-2)}{N}a'^2a^{d-3}\\&+d(d-1)(d-2)kNa^{d-3}-2N\Lambda da^{d-1}-2d(d-1)\frac{d}{d\tau}\left(\frac{a'a^{d-2}}{N}\right)\Bigg)\Bigg]\delta a\,.\end{aligned} \tag{1.18}$$

We thus get the following equation of motion

$$\begin{aligned}&\frac{da^{d-1}}{2N}(\partial_\tau\chi)^2-dNa^{d-1}V_{\rm np}(\chi)+\frac{1}{2\kappa}\Bigg(\frac{d(d-1)(d-2)}{N}a'^2a^{d-3}\\&+d(d-1)(d-2)kNa^{d-3}-2N\Lambda da^{d-1}-2d(d-1)\frac{d}{d\tau}\left(\frac{a'a^{d-2}}{N}\right)\Bigg)=0\,.\end{aligned} \tag{1.19}$$

Finally, varying wrt to N we generate a third (non-independent) equations

$$a^dV_{\rm np}(\chi)+\frac{a^d}{2N^2}(\partial_\tau\chi)^2-\frac{1}{2\kappa}\left[d(d-1)ka^{d-2}-2\Lambda a^d-\frac{d(d-1)a'^2a^{d-2}}{N^2}\right]=0\,. \tag{1.20}$$

Let's summurize the set of equations here for general (k, Λ, N)

$$\begin{aligned}\partial_\tau\left(\frac{a^d}{N}\partial_\tau\chi\right)+Na^d\partial_\chi V_{np}(\chi)&=0\,.\\ \frac{d(d-1)}{a^2}\left(\frac{a'^2}{N^2}-k\right)+2\kappa\left(\frac{(\partial_\tau\chi)^2}{2N^2}+V_{\rm np}(\chi)+\frac{\Lambda}{\kappa}\right)&=0\,,\\ \frac{(d-1)(d-2)}{a^2}\left(k+\frac{a'^2}{N^2}\right)-\frac{2(d-1)}{Na^{d-1}}\frac{d}{d\tau}\left(\frac{a'a^{d-2}}{N}\right)+2\kappa\left(\frac{(\partial_\tau\chi)^2}{2N^2}-V_{\rm np}(\chi)-\frac{\Lambda}{\kappa}\right)&=0\,.\end{aligned} \tag{1.21}$$

Non-running solutions A trivial set of solutions can be obtained by setting χ to a extremum of the potential. These solutions will benchmark of the more general solutions. Let $V_{\rm np}^{\rm ext}$ an extremum of the non-perturbative potential. The geometry of the solution depends crucially on the sign of $\Lambda_{\rm eff}=\kappa V_{\rm np}^{\rm ext}+\Lambda$.

- If $\Lambda_{\rm eff}>0$
  - If $k=1\Rightarrow a(\tau)=\frac{1}{H_{dS}}\sin(H_{dS}\tau)$, with $H_{dS}=\sqrt{\frac{2\Lambda_{\rm eff}}{d(d-1)}}\Rightarrow EdS_{d+1}$ in spherical slices.
  - Otherwise, there are no solutions for a real.
- If $\Lambda_{\rm eff}<0$
  - If $k=1\Rightarrow a(\tau)=\frac{1}{H_{AdS}}\sinh(H_{AdS}\tau)$, with $H_{AdS}=\sqrt{-\frac{2\Lambda_{\rm eff}}{d(d-1)}}\Rightarrow EAdS_{d+1}$ in spherical slices.
  - If $k=-1\Rightarrow a(\tau)=\frac{1}{H_{AdS}}\cosh(H_{AdS}\tau)\Rightarrow EAdS_{d+1}$ in hyperbolic slices.
  - If $k=0\Rightarrow a(\tau)=\frac{1}{H_{AdS}}\exp(H_{AdS}\tau)\Rightarrow EAdS_{d+1}$ in flat slices.
- If $\Lambda_{\rm eff}=0$
  - If $k=1\Rightarrow a(\tau)=\tau\Rightarrow\mathbb{R}_{d+1}$. These are cylinder coordinates associated to Euclidean space.

– If $k = 0 \Rightarrow a(\tau) = \text{const}$. Upon rescaling the transverse flat slices, these are standard cartesian coordinates.

We now focus on compact=spherical slices to obtain a finite action. We could also regulate the infinite volume of the transverse slices but we postpone this issue for latter. In these non-running solutions, the boundary term from the axion part trivially vanishes. We also observe that the bulk on-shell action always take the simple form

$$S_{\text{bulk}}^{\text{on-shell}} = -\frac{2V_d\Lambda_{\text{eff}}}{\kappa(D-2)}\int d\tau a^d \,. \tag{1.22}$$

- If $\Lambda_{\text{eff}} > 0$: There is no boundary since the geometry is a sphere and thus the Gibbons-Hawking York term trivially vanishes. One finds

$$\begin{aligned} S^{\text{on-shell}} &= -\frac{V_d\Lambda_{\text{eff}}}{\kappa(D-2)H_{dS}^{d+1}}\frac{\sqrt{\pi}\Gamma\left(\frac{1+d}{2}\right)}{\Gamma(1+\frac{d}{2})} \\ &= -\frac{d\,V_{d+1}}{\kappa H_{dS}^{d-1}} \,. \end{aligned} \tag{1.23}$$

  This should be seen as the semi-classical contribution to the dS free energy.

- If $\Lambda_{\text{eff}} < 0$: We have an asymptotically EAdS space and thus the action should diverge when approaching the boundary at $\tau \to \infty$. We thus regulate the action by fixing the radius of the boundary $a(\tau_f) = R$

$$\begin{aligned} S_{\text{bulk}}^{\text{on-shell}} &= -\frac{2V_d\Lambda_{\text{eff}}}{\kappa(D-2)}\int d\tau a^d = -\frac{2V_d\Lambda_{\text{eff}}}{\kappa(D-2)H_{AdS}^{d+1}}\int_0^{H_{AdS}R} du\frac{u^d}{\sqrt{1+u^2}} \,. \\ &= -\frac{2V_d\Lambda_{\text{eff}}}{\kappa(D-2)}\frac{R^{d+1}}{d+1}\,{}_2F_1\left(\frac{1}{2},\frac{1+d}{2},\frac{3+d}{2},-(H_{AdS}R)^2\right) \,. \end{aligned} \tag{1.24}$$

  The Gibbons-Hawking yields

$$S_{GHY} = -\frac{dV_d\sqrt{1+H_{AdS}^2R^2}R^{d-1}}{\kappa} \,. \tag{1.25}$$

  Combining these two contributions yield

$$S^{\text{on-shell}} = -\frac{2V_d\Lambda_{\text{eff}}}{\kappa(D-2)}\frac{R^{d+1}}{d+1}\,{}_2F_1\left(\frac{1}{2},\frac{1+d}{2},\frac{3+d}{2},-(H_{AdS}R)^2\right) - \frac{dV_d\sqrt{1+H_{AdS}^2R^2}R^{d-1}}{\kappa} \,. \tag{1.26}$$

- If $\Lambda_{\text{eff}} = 0$: the bulk on-shell action trivially vanishes. We still need to regularize the integral because the GHY term diverges. We thus obtain

$$S^{\text{on-shell}} = -\frac{dV_dR^{d-1}}{\kappa} \,. \tag{1.27}$$